Rat Behavior Recognition Using Deep Learning and Point-Based Annotation

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Problem Statement

◆Human annotation of rat behavior is labor-intensive and time-consuming ◆Human error and inconsistency can result in unreliable data ◆Inaccurate data can result in unreliable findings

Our goal is to create an automated deep-learning system that accurately identifies rat behaviors in video trials, mitigating the obstacles of human annotation and bolstering the study of rat behaviors.

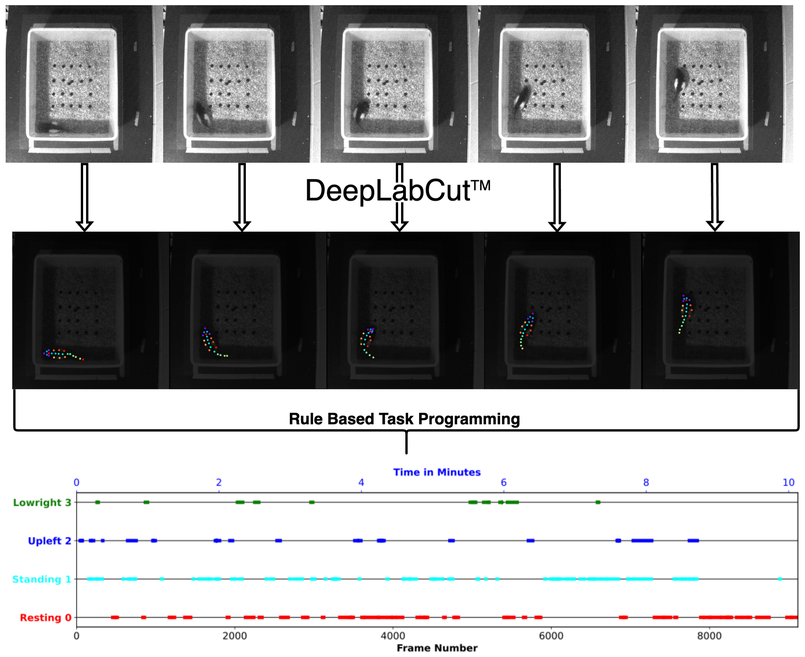
Objectives

◆Train deep learning model to annotate rat key points (e.g. nose, tail) ◆Track rat movement and recognize rule based behaviors ◆Train deep learning model to classify rat behaviors using key points

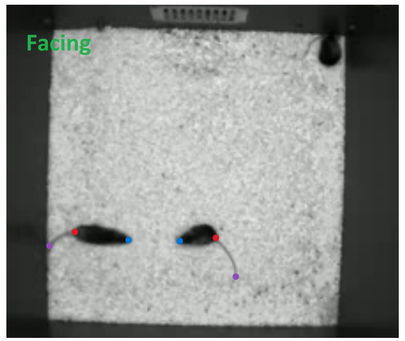
Example Datasets

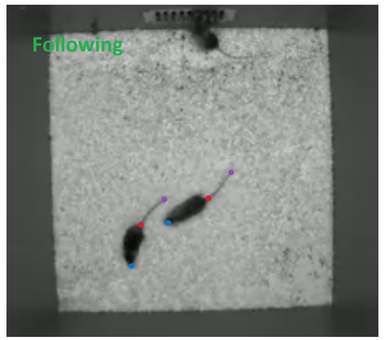
- Marble Interaction
  - Single rat with array of marbles
  - 10-minute videos in gray-scale 480x480 pxs
- Social Exploration
  - Three rats interacting with each other
  - 5-minute videos in gray-scale

Annotation of Marble Interaction Behaviors



Annotation of Social Exploration Behaviors





(Left) Determine whether the two rats are facing each other
(Right) Determine whether one rat is following the other rat
#Images from: J. J. Sun, "The MABe22 Benchmarks for Representation Learning of Multi-Agent Behavior". CaltechDATA, doi: 10.22002/D1.20186.

Rule-based Task Programming (expert-defined for facing)

def: nose_dist $(x_1^{nose}, y_1^{nose}, x_2^{nose}, y_2^{nose})$:

$x_diff = x_2 - x_1$

$y_diff = y_2 - y_1$

return nose_dist = norm(x_diff, y_diff)

def: tail_dist $(x_1^{tail}, y_1^{tail}, x_2^{tail}, y_2^{tail})$:

$x_diff = x_2 - x_1$

$y_diff = y_2 - y_1$

return tail_dist = norm(x_diff, y_diff)

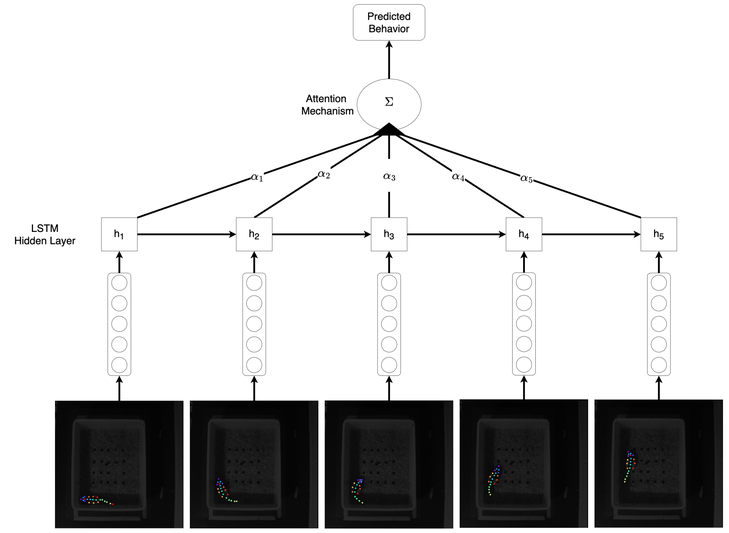
def: is_facing $(\mathbf{X}_1, \mathbf{Y}_1, \mathbf{X}_2, \mathbf{Y}_2,$ nose_dist, tail_dist$)$:

m2m_angle = acos $((\mathbf{X}_1, \mathbf{Y}_1) \cdot (\mathbf{X}_2, \mathbf{Y}_2) \; / \; | (\mathbf{X}_1, \mathbf{Y}_1) | | (\mathbf{X}_2, \mathbf{Y}_2) |)$

return is_facing **if** nose_dist $<$ tail_dist **and** m2m_angle $<$ threshold

Resulting annotations can be manually corrected or used as weak supervision in scarce labeled data settings (e.g., semisupervised learning)

Behavior Classification using deep learning



Ongoing Work

- Use self-supervised learning to learn trajectory representations
- Evaluate few-shot and semi-supervised approaches for behavior classification on pre-trained representations
- Integrate task programming into the learning task (Sun et al., 2021)
- Explore transformer-based models as an alternative to LSTM models
- Investigate the transfer of models from one task to another

Acknowledgements

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